A PROJECT REPORT

on

"Sentiment Classification of IMDb Movie Reviews using Random Forest"

Submitted to

KIIT Deemed to be University

In Partial Fulfilment of the Requirement for the Award of

BACHELOR’S DEGREE IN INFORMATION TECHNOLOGY

BY

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UNDER THE GUIDANCE OF

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April 2025

CERTIFICATE

This is to certify that the project entitled

"Sentiment Classification of IMDb Movie Reviews using Random Forest"

submitted by

ARMAAN GUPTA ROLL NUMBER: 22053846 GROUP NUMBER: 30

is a record of bonafide work carried out by him, in the partial fulfilment of the requirement for the award of Degree of Bachelor of Technology in Information Technology at KIIT Deemed to be University, Bhubaneswar. This work was done during the year 2024-2025, under our guidance.

Date:

(Signature) Ms. Junali Jasmine
Project Guide

ACKNOWLEDGEMENTS

I am profoundly grateful to Ms. Junali Jasmine of the School of Computer Engineering for her expert guidance and continuous encouragement throughout this project. Her valuable feedback and constant support played a crucial role in shaping this work.

I also thank the faculty and staff of the School of Computer Engineering at KIIT for providing a conducive environment for research and development.

ARMAAN GUPTA

ABSTRACT

This project focuses on sentiment classification of IMDb movie reviews using machine learning techniques. The primary goal is to distinguish between positive and negative reviews to enhance automated opinion mining. A dataset of 50,000 IMDb reviews was preprocessed and used to train a Random Forest classifier, which achieved promising accuracy. Various stages including data cleaning, feature extraction with n-grams, and evaluation were undertaken.

Keywords: Sentiment Analysis, IMDb, Machine Learning, Random Forest, Text Classification

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Chapter 1: Introduction

Sentiment Analysis, also known as opinion mining, is one of the most actively researched areas in Natural Language Processing. With the explosive growth of online platforms, users are constantly reviewing content, especially movies. Identifying whether a review is positive or negative holds significant value for movie producers, advertisers, and audiences.

The IMDb dataset consists of 50,000 labeled movie reviews, making it an ideal choice for training supervised learning models. This project aims to implement an effective sentiment classifier using a Random Forest algorithm. The report is structured to walk through the methodology starting from preprocessing to model deployment.

Chapter 2: Basic Concepts / Literature Review

2.1 Natural Language Processing (NLP)

NLP enables machines to understand, interpret, and manipulate human language. Techniques such as tokenization, stopword removal, and stemming are used for preprocessing text.

2.2 Bag-of-Words & N-grams

These are text representation techniques. Bag-of-Words captures word frequency, while n-grams capture local context (e.g., pairs of consecutive words).

2.3 Random Forest Classifier

An ensemble learning method that builds multiple decision trees and combines their outputs to improve classification accuracy and reduce overfitting.

Chapter 3: Problem Statement / Requirement Specifications

The goal is to develop a classifier that can predict the sentiment of movie reviews from IMDb.

3.1 Project Planning

- Data Collection (IMDb dataset)
- Preprocessing (cleaning, stopword removal)
- Feature Extraction (BoW with n-grams)
- Model Training (Random Forest)
- Evaluation and Testing

3.2 Project Analysis

Care was taken to balance classes, handle duplicates, and ensure valid test splits. Ambiguity was resolved through repeated evaluation and experimentation.

3.3 System Design

3.3.1 Design Constraints:

- Language: Python
- Tools: Google Colab
- Libraries: pandas, scikit-learn, nltk
- Hardware: Standard Colab runtime with GPU/CPU

3.3.2 System Architecture:

A simple ML pipeline from preprocessing to vectorization and classification using BoW and Random Forest.

Chapter 4: Implementation

4.1 Methodology

- Load dataset using `kagglehub`
- Clean HTML tags using regex
- Lowercase conversion, stopword removal using `nltk`
- Feature extraction using `CountVectorizer` with unigrams and bigrams
- Model training with `RandomForestClassifier`

4.2 Testing Plan

Testing was done using the test set split (80:20). Accuracy and confusion matrix were computed.

4.3 Result Analysis

The model achieved consistent performance with accuracy close to 85%. It was also tested on a custom sentence and responded accurately.

4.4 Quality Assurance

Best coding practices and sklearn standard functions were used for model training and testing. Pickle was used to save the model and vectorizer.

Chapter 5: Standards Adopted

5.1 Design Standards

Followed scikit-learn and Python design patterns and modularization of code.

5.2 Coding Standards

- Function-based structure
- Proper variable naming
- Inline comments
- Avoided long functions

5.3 Testing Standards

- Accuracy score
- Confusion matrix evaluation
- Manual testing for individual predictions

Chapter 6: Conclusion and Future Scope

6.1 Conclusion

The project successfully demonstrated sentiment classification using machine learning. The Random Forest classifier showed effective performance on IMDb movie reviews.

6.2 Future Scope

Future work can include:

- Incorporating deep learning models like LSTMs or transformers
- Expanding dataset size and language support
- Real-time API-based sentiment analysis

References

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